

In-Class Exercise Sheet 2

1. Let's discuss the article "Reproducibility with VisTrails".

(a) Name the levels of reproducibility, filling in the blanks:

- _____, how much of an experiment is available,
- _____, whether the experiment is repeatable, reproducible, or even replicable, and
- _____, how much of the experiment can be confirmed.

For instance, considering an experiment that requires special hardware to derive data,

_____ reproducibility can be obtained by providing the data produced by the hardware, and the analysis processes to derive the plots from the paper.

(b) Name the types of provenance, relevant for reproducibility:

- _____ provenance embodies the description of an experiment, the specification of the workflow structure, including modules, connections, and inputs.
- _____ provenance captures information about the execution of the workflow, what actually happened when the workflow was run.
- _____ captures the history of the workflow, that is all the different versions of the workflow.

(c) As of today, is the VisTrails project still being maintained?

☐ Yes ☐ No

What is your conclusion?

2. We discuss aspects of the *Nature* article “Reproducibility standards for machine learning in the life sciences” by Heil et al., published in 2021.

- (a) A main motivation stated in the article is:

“For machine-learning models in the life sciences to become _____, scientists must prioritize computational reproducibility.”

- (b) The article proposes different reproducibility standards. Fill in the blanks in the table.

	Bronze	Silver	Gold
Data published and downloadable			
Models published and downloadable			
Source code published and downloadable			
Dependencies set up in a single command			
Key analysis details recorded			
Analysis components set to deterministic			
Entire analysis reproducible with a single command			

- (c) What problem is portrayed in the article for the case where the authors merely *report* on their experiments and their model?

- (d) The article proposes several platforms for sharing research artefacts, among the general-purpose platforms are

1. _____ for datasets of up to 50GB and
2. _____ for datasets larger than 50GB.

- (e) One best practice recommended in the article is to create a GitHub project and archive it in Zenodo. Why twice the effort?

- (f) The article points out a hardware-related problem in the context of machine learning. Which is it?

- (g) What is “badging” and why is it an incentive for making research reproducible?

- (h) What recommendation is given for compute-intensive analyses?

3. Describing methods and results. Insert the words below into the spaces.

following, having, reducing, resulting, selecting, speaking, subtracting, taking, using

- (a) The samples were prepared _____ Jude [2012].
- (b) The third mixture was prepared _____ the same procedure as for the first.
- (c) The criteria for _____ the subjects to participate in the survey were: ...
- (d) By _____ the amount of liquid to the minimum, the mixture becomes more solid.
- (e) Generally _____, our results show that bankers have no social conscience.

- (f) The _____ solution to this problem can be expressed as ...
- (g) _____ advantage of the properties of gold, we can now ...
- (h) _____ the first result from the second, we obtain ...
- (i) _____ these features meant that we could ...

Exercise adapted from Adrian Wallwork's book on "English for Academic Research: Vocabulary Exercises", published at Springer.

4. **Highlighting drawbacks of previous research.** Insert a suitable word from the list below into the spaces.

appropriate, complicated, concern, conjectures, drawback, flawed, misleading, shortcomings, speculative, weakness

- (a) The _____ of their method have been clearly recognized.
- (b) A serious _____ with this argument, however, is that ...
- (c) Their approach is not _____ for those kinds of patients.
- (d) Their experiments were _____ by the fact that they were almost impossible to replicate.
- (e) This then is the major _____ to their experiments.
- (f) Such a _____ assumption can lead to serious consequences with regard to ...
- (g) Their claims seem to be somewhat _____.
- (h) In our view, their findings are only _____ based on unsubstantiated assumptions.
- (i) Their attempts to solve this simple problem are unnecessarily _____.
- (j) An even greater source of _____ is the fact that ...

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